

An Overview of the Workforce Implications of Artificial Intelligence

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The beginning of a new kind of software is here

- Newly unlocked is the ability to analyze unstructured data (images, video, text, code, sequences...)
- The results have already been transformative. Occasionally eerie as well.
- Business value lives in both the “encoder” and “decoder” components of these models.
- **Machines can output knowledge.**
- What is AI now... what was it? What do you use it for?

Something regression can't do... what's this?



How can we get a model to do that?





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81	49	31	73	55	79	14	29	93	71	40	67	58	88	30	03	49	13	36	65
52	70	95	23	04	60	11	42	68	44	68	56	01	32	56	71	37	02	36	91
22	31	16	71	51	67	83	59	41	92	36	54	22	40	40	28	66	33	13	80
24	47	37	00	99	03	45	02	44	75	33	53	78	36	84	20	35	17	12	50
52	98	81	28	64	23	67	10	26	38	40	67	59	54	70	66	18	38	64	70
67	26	20	68	02	62	12	20	95	63	94	39	63	08	40	91	66	49	94	21
24	55	58	05	66	73	99	26	97	17	78	78	96	83	14	88	34	69	63	72
21	36	23	09	75	00	76	44	20	45	35	14	00	61	33	97	34	31	33	95
78	17	53	28	22	75	31	67	15	94	03	80	04	62	16	14	09	53	56	92
16	39	05	42	96	35	31	47	55	58	88	24	00	17	54	24	36	29	85	57
86	56	00	48	35	71	89	07	05	44	44	37	44	60	21	58	51	54	17	58
19	80	81	68	05	94	47	69	28	73	92	13	86	52	17	77	04	89	55	40
04	52	08	83	97	35	99	16	07	97	57	32	16	26	26	79	33	27	98	66
55	46	68	87	57	62	20	72	03	46	33	67	46	55	12	32	63	93	53	69
04	42	16	73	25	39	11	24	94	72	18	08	46	29	32	40	62	76	36	36
20	69	36	41	72	30	23	88	31	58	89	82	67	59	85	74	04	36	16	16
20	73	35	29	78	31	90	01	74	31	49	71	48	58	81	16	23	57	05	54
01	70	54	71	83	51	54	69	16	92	33	48	61	43	52	01	89	79	62	48

What the computer sees

image classification

82% cat
15% dog
2% hat
1% mug

■ **"Pure" Reinforcement Learning (cherry)**

- ▶ The machine predicts a scalar reward given once in a while.
- ▶ **A few bits for some samples**

■ **Supervised Learning (icing)**

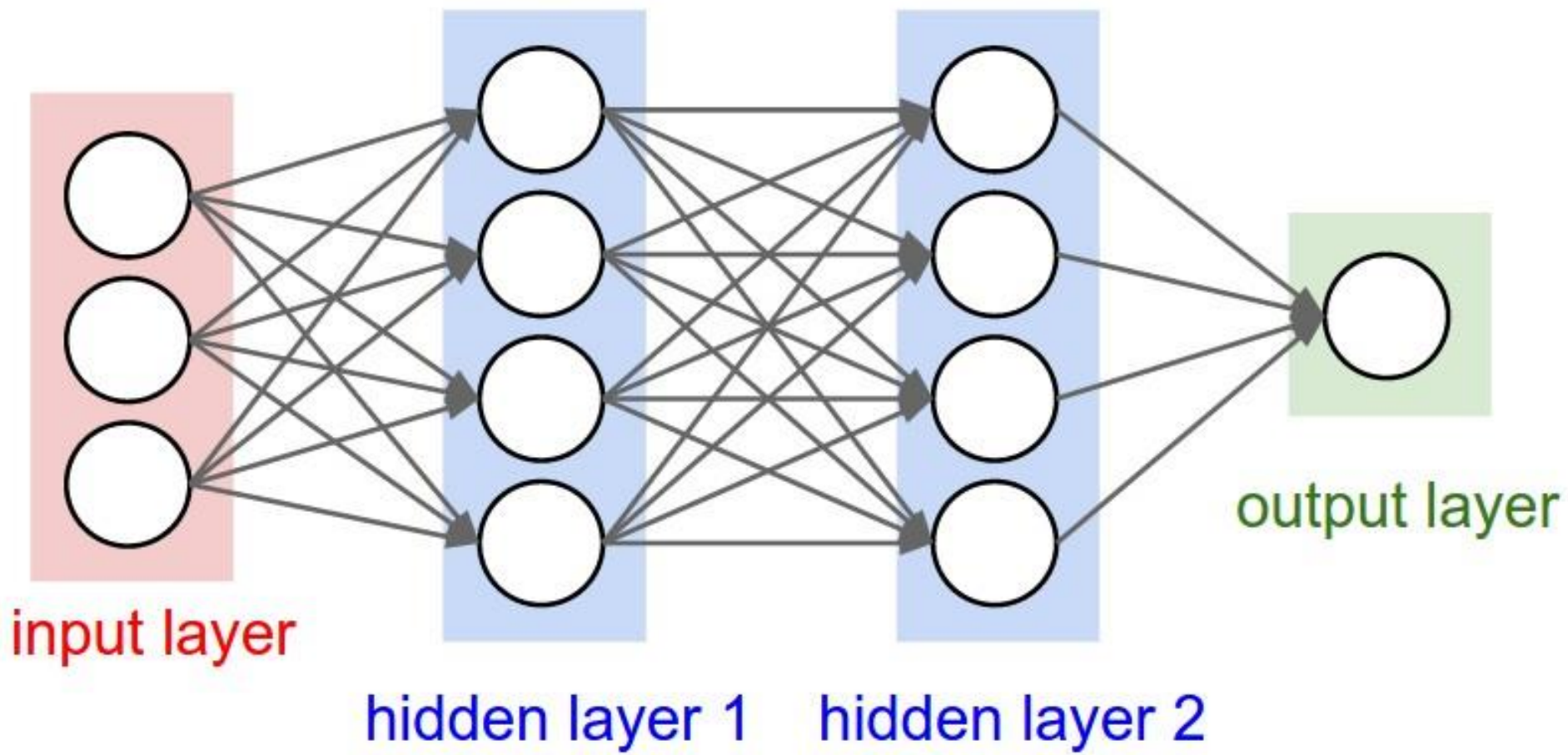
- ▶ The machine predicts a category or a few numbers for each input
- ▶ Predicting human-supplied data
- ▶ **10→10,000 bits per sample**

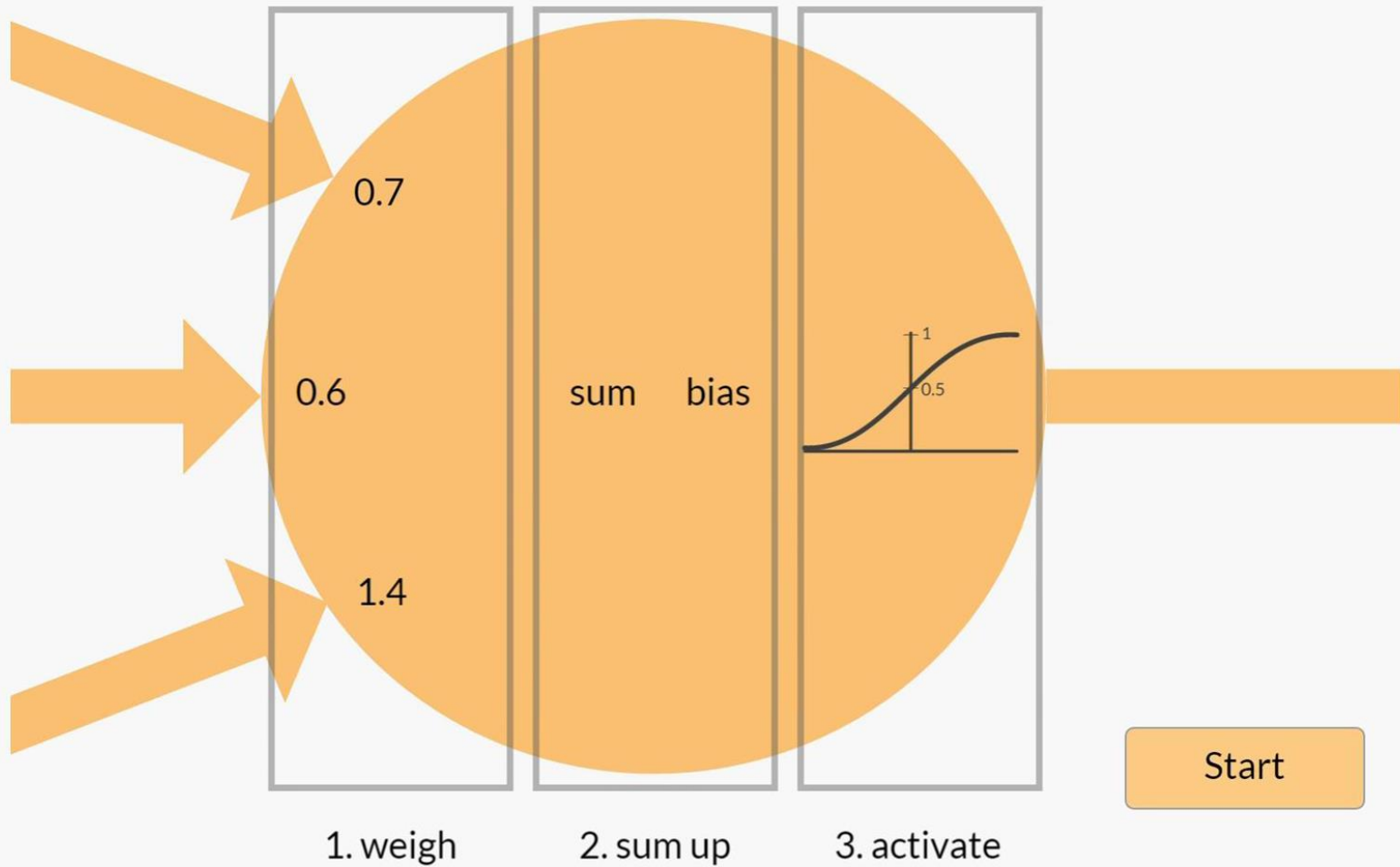
■ **Unsupervised/Predictive Learning (cake)**

- ▶ The machine predicts any part of its input for any observed part.
- ▶ Predicts future frames in videos
- ▶ **Millions of bits per sample**

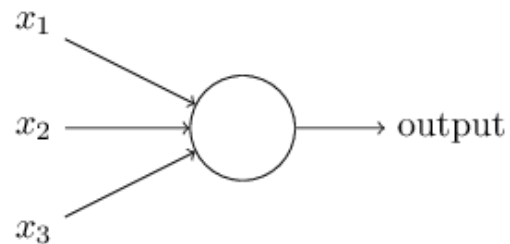


■ (Yes, I know, this picture is slightly offensive to RL folks. But I'll make it up)

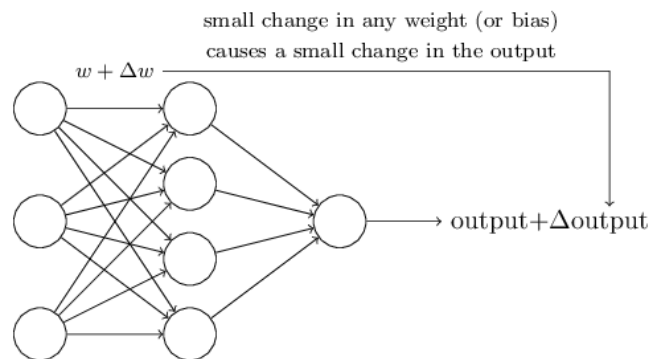




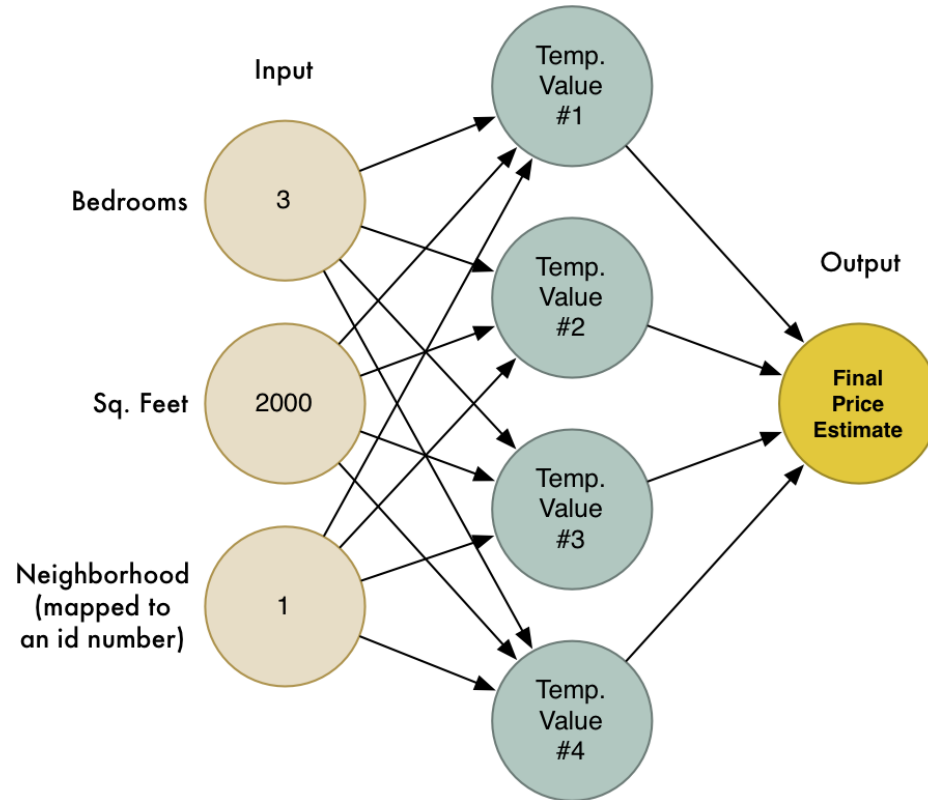
The perceptron as transformer



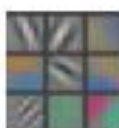
$$\text{output} = \begin{cases} 0 & \text{if } \sum_j w_j x_j \leq \text{threshold} \\ 1 & \text{if } \sum_j w_j x_j > \text{threshold} \end{cases}$$



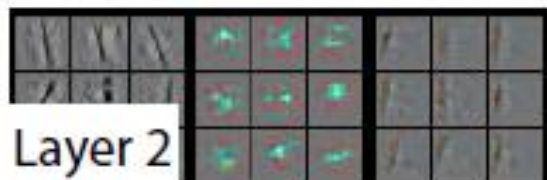
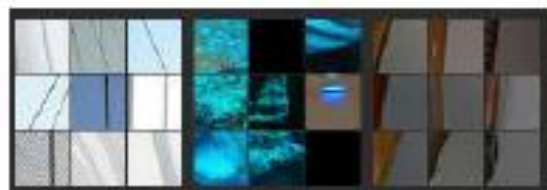
What do the nodes mean?







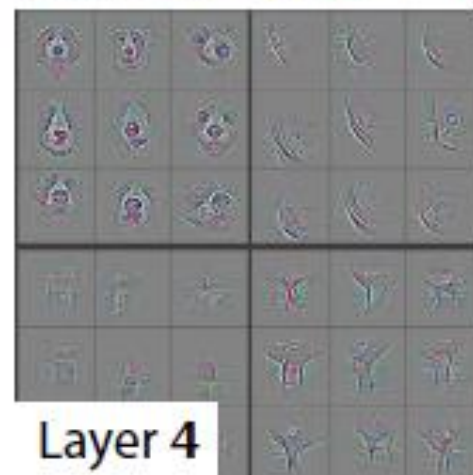
Layer 1



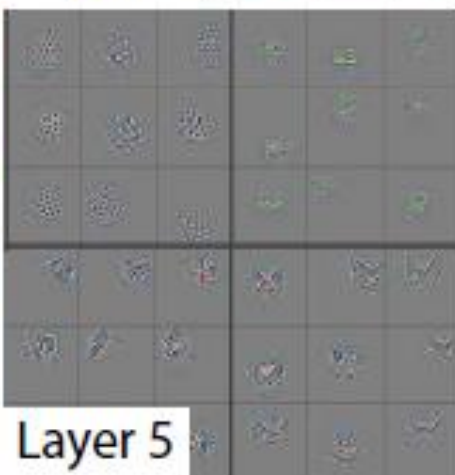
Layer 2



Layer 3



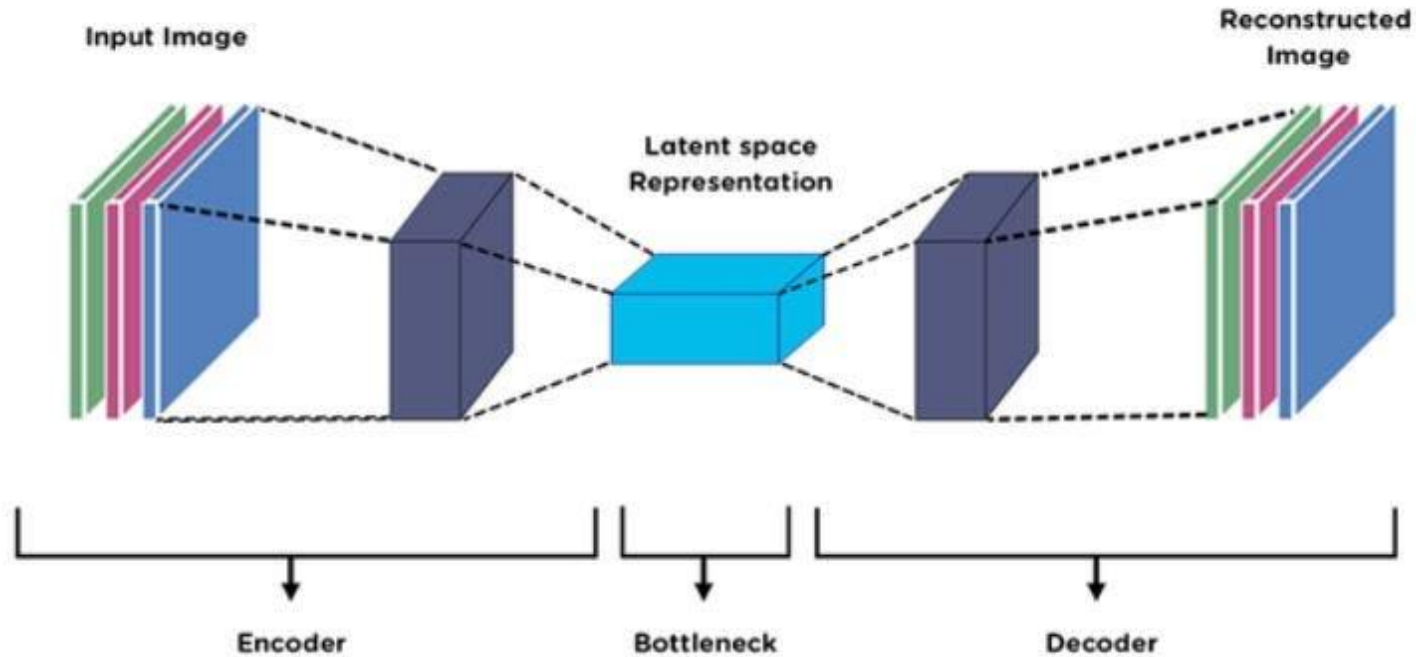
Layer 4



Layer 5



The beginning of a new kind of software is here



Generative AI will be transformative and will require a plan.

- This is not the robot apocalypse. It is a massive opportunity.
- Now is **the time to get moving**. Starting an implementation plan is table stakes because this technology is moving quickly. Others are already making strides.
- Of course, “**measure twice, cut once**”. It can be helpful to get some quick wins under your belt first.
- **Set up your platform for success**. What assets do you have now?
- **Data, Talent, and Computation** will take you further now than they have in the past. Make these resources accessible for building new projects.
- Given wide-ranging exposure of knowledge work, **workforce strategy will be critical** in determining how successful generative AI projects will be.

Where we are vs. where we're going



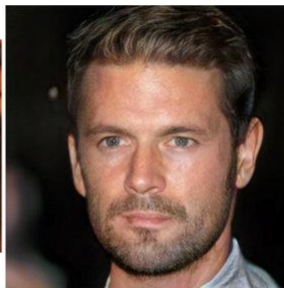
2014



2015



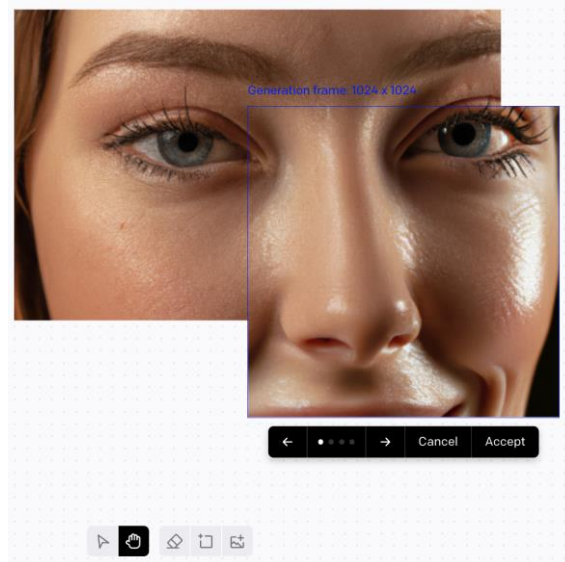
2016



2017

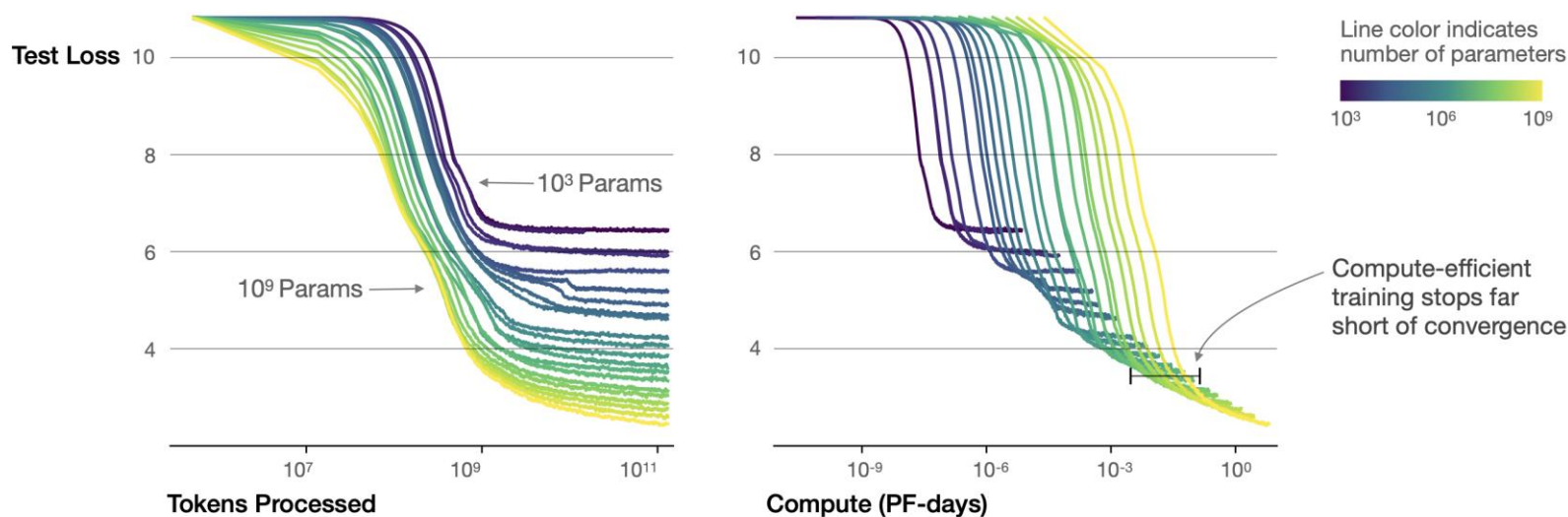


2018



Let's discuss why these technologies have gotten better

- **Scale**
- **Taste**

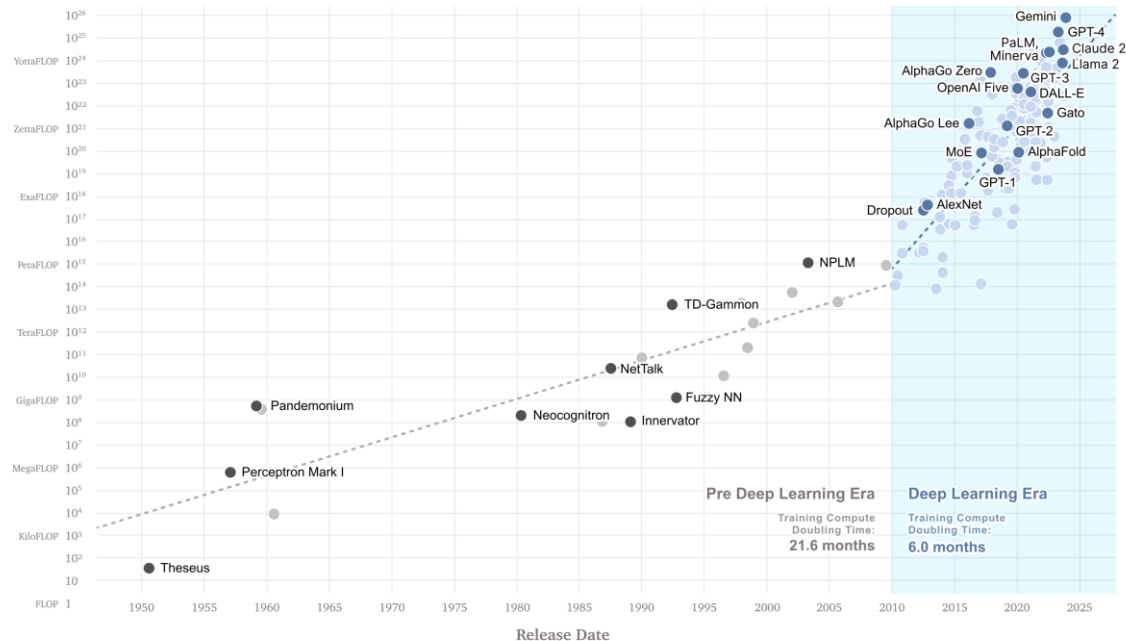


Source: (Kaplan et al., 2020)

Scaling in compute has hit an accelerated stride

Compute Used for AI Training Runs

Total compute used to train notable AI models, measured in total FLOP (floating-point operations) | Logarithmic



(a) Pre-2010 Trend. Compute usage for training AI systems before 2010 doubled every 1.8 months. This tracks Moore's Law-esque improvements in compute price-performance (doubling every two years).

Sastry et al. (2024)

Figure 5: The importance of compute AI in a historical context. (Data from Epoch (2023) and Sevilla, Heim, A. Ho, et al. (2022).)

- Scale
- Taste

USER

Can you translate Taylor Swift's "Shake It Off" lyrics into a song in Led Zeppelin's style?

ASSISTANT

(Verse 1)

I slid my gaze across the yard, tales spreading further far,
To my back resonance of these guys speaking too hard,
They rinse them lips with chronic pints of insipid gumption,
Hollow minds skip rocks on waves of compiled assumption.

A filament guise of hate burns an inscrutable slander wax,
Snap unholy grammar n}most steadily behind our backs.
Yeah, they claim they've seen me acquiring the Venus dance charms,
Mark time - oversight echos fueled graffiti by unaware's balms.

(Amphoric Chorus)

Emancipate emotion reels, beckon sol storm unreals,
Shatter enmity embrace with justify thy posit appeal.
People scattering spite gems, emboss timelines with disgrace ink,
Rev Photo Atlantic rollers - crush Cyclopean cleanse gift ere forever sink, hey!

Oh heal nurture thy heated boon, Thrum each whispered sinful psalm lied.
Merge cosmic cord winding Blues, lead the walking hominid mystics, Terras sighed.

(Overbearing Lydia shuffling, spinning circles solo twee Led prototype)

(Verse 2)

Not fast breeder in espaleigh sick spite glorification tracing morning drought,
Banking on trades tainted nitrogen climb once theoretical zealot event shouts.
Prop hammer-Drum partner trap replay strides close unknown dark,
Imitating badly bridled cosmos energy embarrass phagic rookie mark.

(Acoustic guitaristarium excitement jim restructuring Myers delt, ey!)

(Amphoric Chorus)

Synchronic cloud weave, unabashed, decimate distortion dreadful tease,
Speeding hollow veneer train embellished presumed tiles toxic nucleus keys.

- Scale
- Taste

ASSISTANT

(Verse 1)

In the light of the morning, I hear the whispers start,
They're spinning tales about me, trying to break my heart.
They say I'm dancing with strangers, lost in the city lights,
But I'm just finding my rhythm, in the midst of these lonely nights.

(Chorus)

So I'll shake it off, shake it off, let the rumors fly,
I'll shake it off, shake it off, under the endless sky.
They can't bring me down, with their words so cold,
I'll shake it off, shake it off, let the truth unfold.

(Verse 2)

They're painting pictures about me, with colors of deceit,
But I won't let their shadows, knock me off my feet.
I'm dancing to my own beat, under the moon's soft glow,
Their words can't touch me, I'm letting go.

(Chorus)

So I'll shake it off, shake it off, let the rumors fly,
I'll shake it off, shake it off, under the endless sky.
They can't bring me down, with their words so cold,
I'll shake it off, shake it off, let the truth unfold.

General-Purpose Technology Criteria:

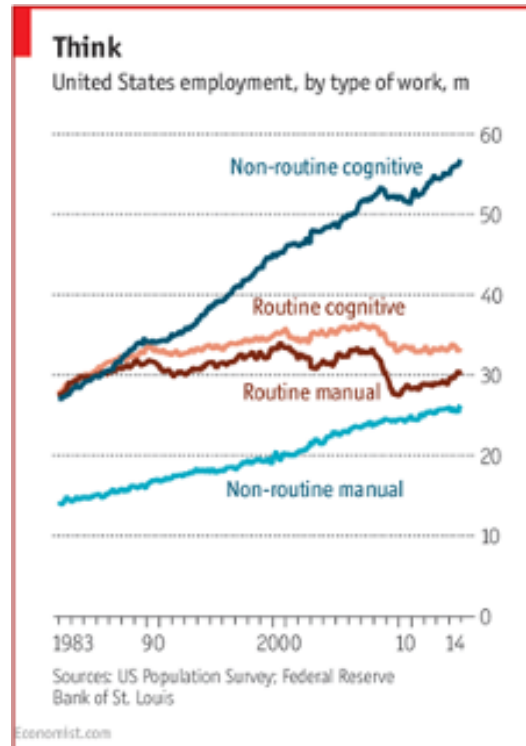
- Pervasive
 - Check: Do lots of occupations have exposure?
- Improves over time
 - Check: Developer activity, model improvements, we're going to take it for granted too
- Spawns complementary innovation
 - Check: Is occupational exposure also contingent on building with other systems?

Workforce Considerations

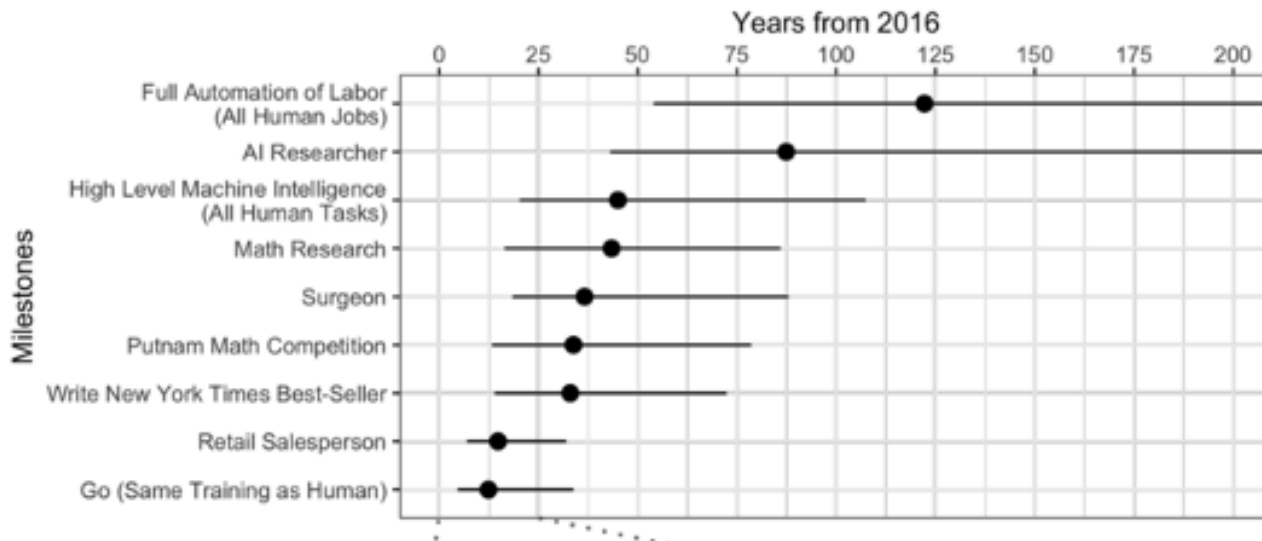
The quick version of workforce impacts to expect

- Around 80 percent of workers have at least 10 percent of their tasks exposed
 - While workers in some tasks might immediately become more productive, much of the value is locked up: **additional tools and software unlock the value of AI**
 - Exposure is neither good nor bad! It means we can expect change.
- Exposure to generative AI is pervasive across industries and occupations
 - The types of work generative AI helps with typically relate to **information processing**.
 - Jobs probably aren't the right unit of analysis. Consider **tasks** and **systems**.

IT job displacement arguments aren't new



Why is this time different? Is it different? The Most 'G' of all GPTs.



Task ID	Occupation Title	DWAs	Task Description
14675	Computer Systems Engineers/Architects	Monitor computer system performance to ensure proper operation.	Monitor system operation to detect potential problems.
18310	Acute Care Nurses	Operate diagnostic or therapeutic medical instruments or equipment. Prepare medical supplies or equipment for use.	Set up, operate, or monitor invasive equipment and devices, such as colostomy or tracheotomy equipment, mechanical ventilators, catheters, gastrointestinal tubes, and central lines.
4668.0	Gambling Cage Workers	Execute sales or other financial transactions.	Cash checks and process credit card advances for patrons.
15709	Online Merchants	Execute sales or other financial transactions.	Deliver e-mail confirmation of completed transactions and shipment.
6529	Kindergarten Teachers, Except Special Education	–	Involve parent volunteers and older students in children’s activities to facilitate involvement in focused, complex play.
6568	Elementary School Teachers, Except Special Education	–	Involve parent volunteers and older students in children’s activities to facilitate involvement in focused, complex play.

Table 1: Sample of occupations, tasks, and Detailed Work Activities from the O*NET database. We see that aggregating over activities alone is imprecise, as evidenced by the fact that we’d expect Gambling Cage Workers to complete the given DWA in person, using some physicality while we’d expect Online Merchants to complete the same activity solely with a computer.

Summary of exposure rubric

No exposure (E0) if:

- using the described LLM results in no or minimal reduction in the time required to complete the activity or task while maintaining equivalent quality^a or
- using the described LLM results in a decrease in the quality of the activity/task output.

Direct exposure (E1) if:

- using the described LLM via ChatGPT or the OpenAI playground can decrease the time required to complete the DWA or task by at least half (50%).

LLM+ Exposed (E2) if:

- access to the described LLM alone would not reduce the time required to complete the activity/task by at least half, but
- additional software could be developed on top of the LLM that could reduce the time it takes to complete the specific activity/task with quality by at least half. Among these systems, we count access to image generation systems.^b

^aEquivalent quality means that a third party, typically the recipient of the output, would not notice or care about LLM assistance.

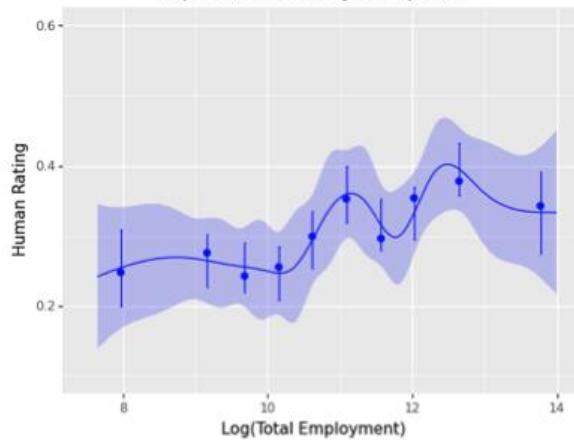
^bIn practice, as can be seen in the full rubric in Appendix A.1, we categorize access to image capabilities separately (E3) to facilitate annotation, though we combine E2 and E3 for all analyses.

What are the most exposed roles? They tend to be knowledge workers.

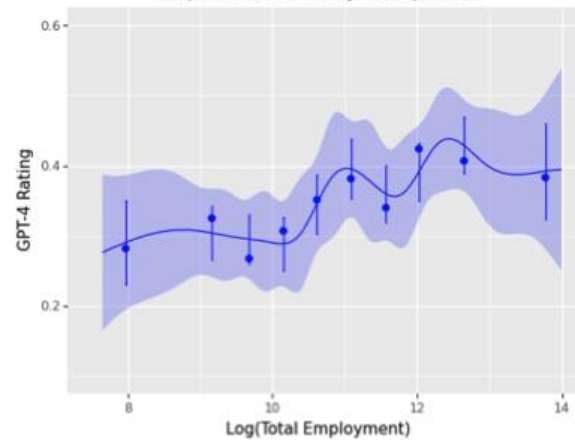
Title
Mathematicians
Proofreaders and Copy Markers
Blockchain Engineers
Correspondence Clerks
Court Reporters and Simultaneous Captioners
Computer Programmers
Data Warehousing Specialists
Database Administrators
Web Developers
Web Administrators
Telephone Operators
Data Entry Keyers
Statistical Assistants
Poets, Lyricists and Creative Writers
Interpreters and Translators
Database Architects
Software Quality Assurance Analysts and Testers
Medical Transcriptionists
Software Developers
Bioinformatics Technicians

- These technologies are triggering a repricing of human capital.
- This is a new kind of software, opening up frontiers in generating value from data.

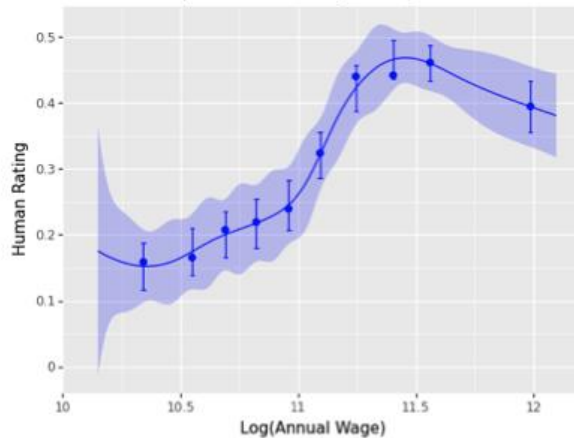
Exposure to GPTs by Occupation



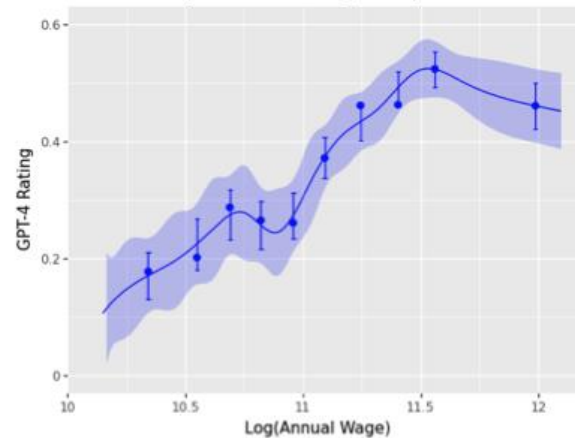
Exposure to GPTs by Occupation



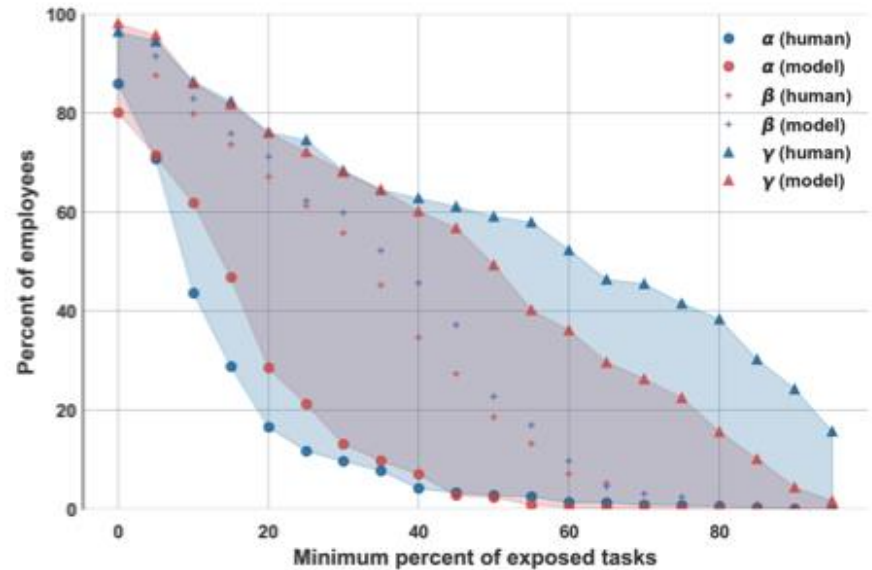
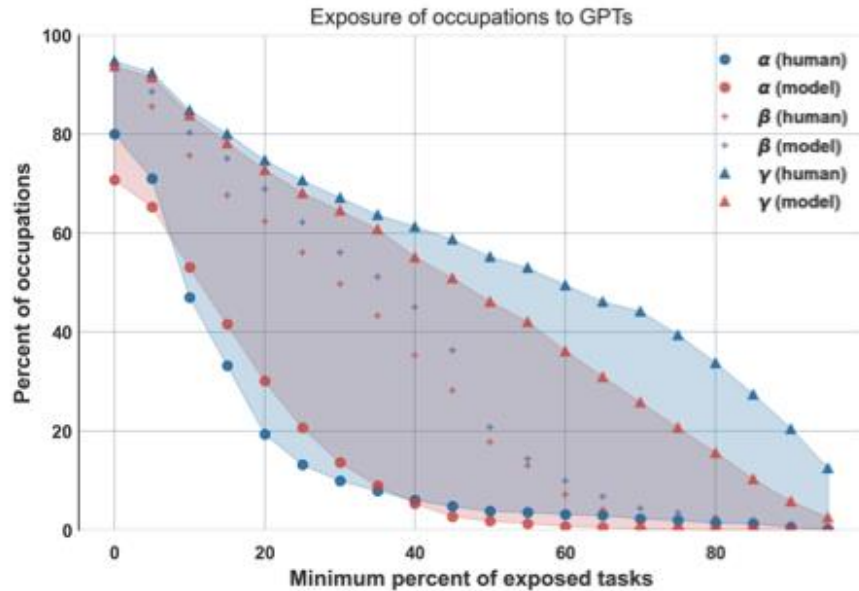
Exposure to GPTs by Occupation



Exposure to GPTs by Occupation



Unlocking AI potential means building innovative tools, software, and processes



More exposed roles typically have greater barriers to entry

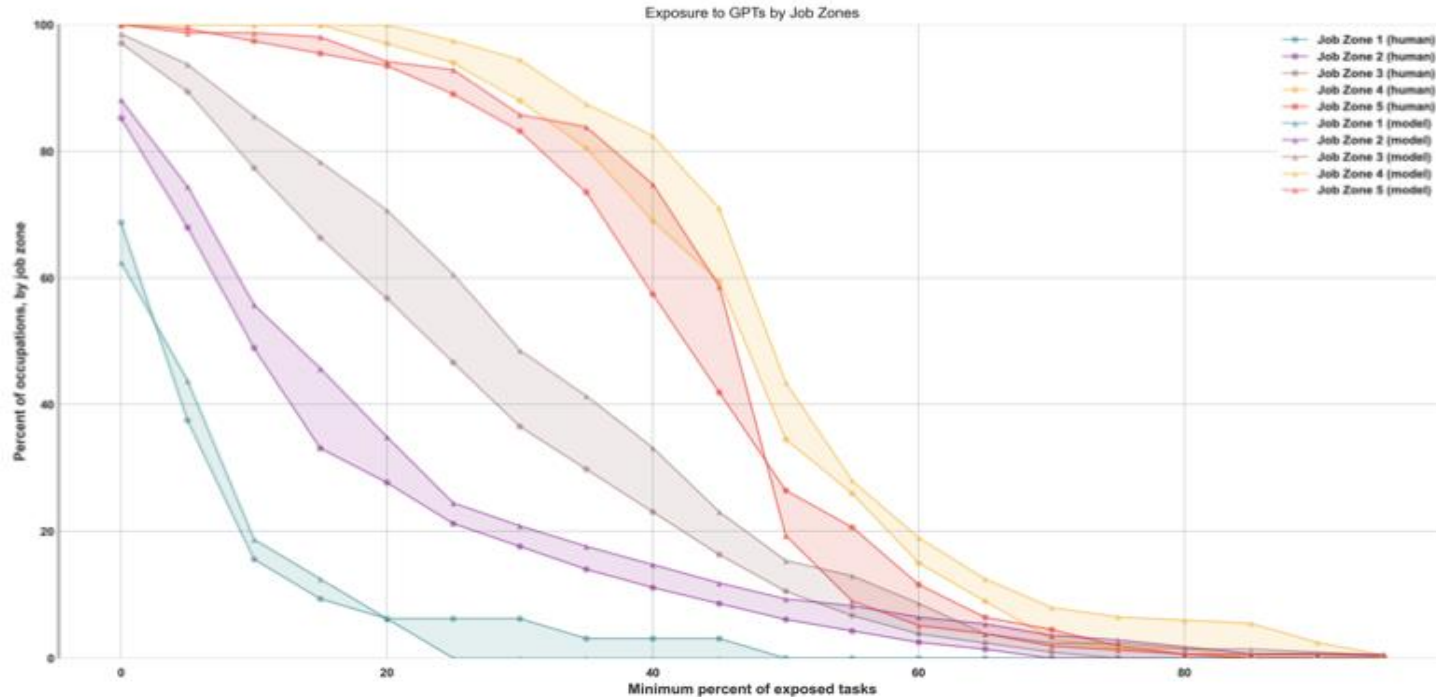
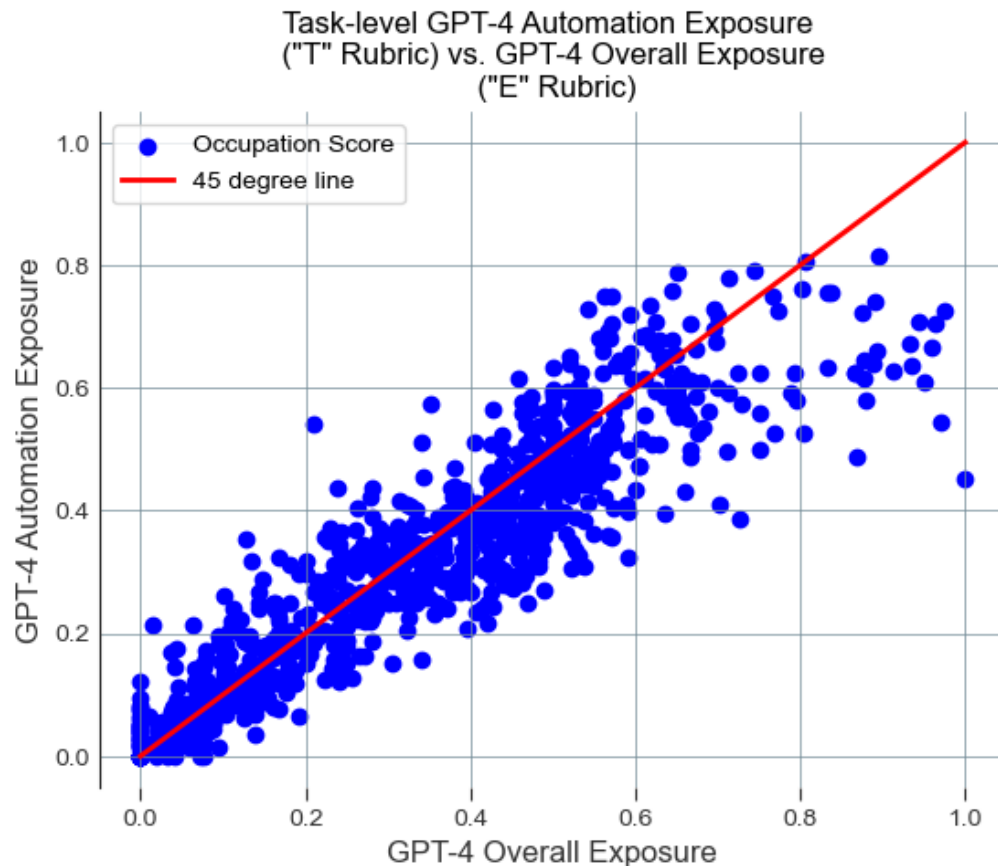


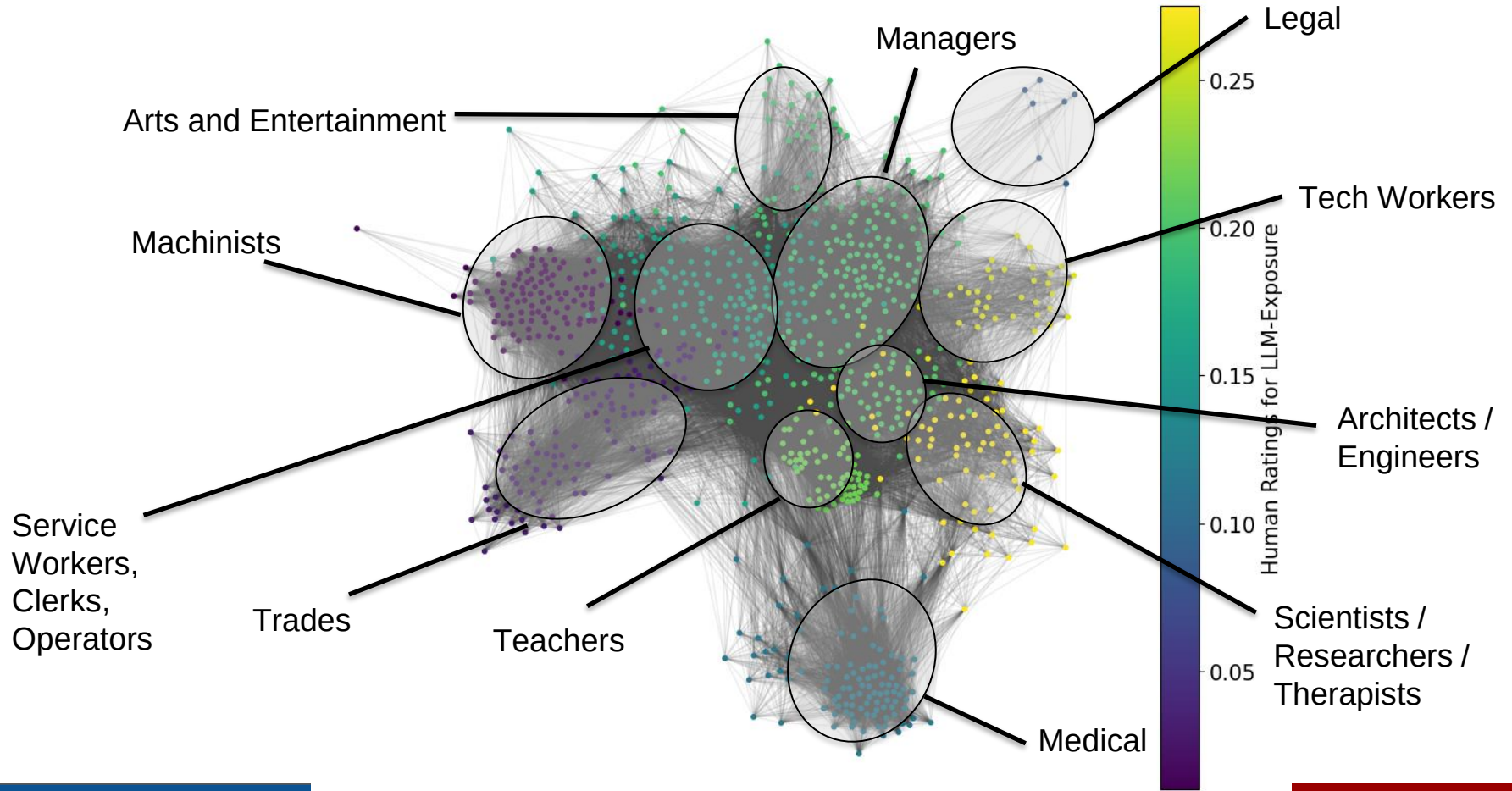
Figure 5: β exposure ratings of occupations in the five Job Zones, which are groups of similar occupations that are classified according to the level of education, experience, and on-the-job training needed to perform them.

What about automation?

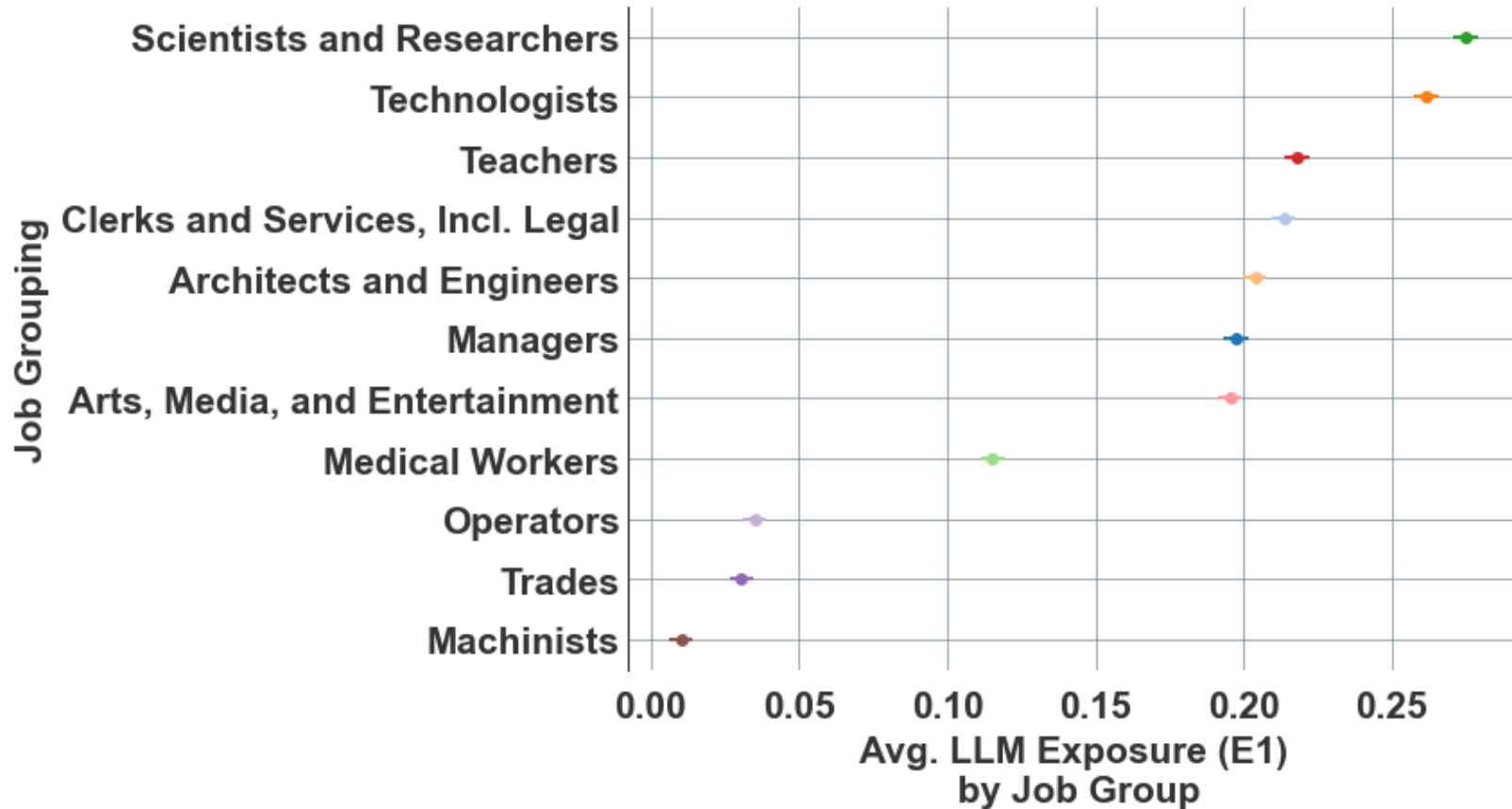
Title
Telephone Operators
Telemarketers
Word Processors and Typists
Credit Authorizers, Checkers, and Clerks
Order Clerks
Bookkeeping, Accounting, and Auditing Clerks
Desktop Publishers
Payroll and Timekeeping Clerks
Insurance Claims and Policy Processing Clerks
Brokerage Clerks
Insurance Underwriters
Travel Agents
Statistical Assistants
Medical Records Specialists
Tellers
Legal Secretaries and Administrative Assistants
Billing and Posting Clerks
Proofreaders and Copy Markers
Medical Transcriptionists
Loan Interviewers and Clerks



Researchers and developers rank amongst the most exposed groups



Clustering exposure (just to LLMs) by job "archetypes" shows the pattern



There are many important caveats to this analysis (internal to the study)

- **Subjective human judgments:** Labelers understand LLM capabilities, but don't know these roles deeply.
- **Measuring LLMs with GPT-4:** Brittle rubrics and arbitrary thresholds. Iteration leads to slightly different results.
- **Validity of the task-based framework:** What is the atomic unit of work? These task lists are one instrument with many imperfections. Some tasks are {up, down}stream of others. Big assumption here that this dataset reflects some organization of work.
- **Lack of human annotator expertise:** Annotators unaware of occupations $\leftrightarrow$ activities. (Collected more labels in some cases)
- **Forward-looking and subject to change:** This is an ongoing effort and the equilibrium is very hard to predict.
- **Disagreement between humans and GPT-4:** Humans and GPT-4 are differentially aware of context. This can change results and makes outputs of the model sensitive to prompting (among other concerns).
- **Saying nothing about social, legal/regulatory, political considerations:** Technical feasibility is only one part of the process.
- **Arbitrary focus on software:** Robots are starting to use LLMs...

Let's talk about telephone switchboard operators.

(Feigenbaum and Gross 2020)

- Telephone operator work was mostly automated from 1920-1940
- But didn't affect future worker cohorts. They found other jobs.
- It *did* negatively impact incumbent workers.

Who are the telephone operators of today?

Where will they go?



What tasks are in demand in the modern economy?

What skills and qualifications are employers looking for?

What are they willing to pay for them?

Switchboard Operator

Wilmington, DE

Apply

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A customer service agent who is responsible to operate telephone switchboard using computer and headset. Expedite incoming, outgoing and interoffice calls. Maintain daily on call information via computer for paging medical staff and other hospital personnel as required.

Essential Functions:

- Answer calls as they appear on the console and assist caller in locating the requested associate or department.
- Maintain pleasant, courteous and calm attitude throughout the duration of all calls.
- Fully assess need of caller and transfer call in the most expeditious manner to the intended associate or department.
- Utilize on-line, on-call schedules to identify correct on-call medical staff and other key administrative personnel.
- Answer Private Physician Line and make necessary pages as needed
- Report all telephone console problems to the Helpdesk for resolution.
- Adhere to key performance indicators and identify opportunities for continuous improvement.
- Maintains confidentiality at all times.
- Adhere to Nemours policies, procedures, and practice the core values and Standards of Behavior of the hospital.

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Machine Learning, Jobs and Wages

- How Does Technology Affects Wages?
 1. Substitution

There are (at least) six forces at work

- What Affects Wages?
 1. Substitution
 2. Complementarities
 3. Demand Elasticity
 4. Income Elasticity
 5. Supply Elasticity
 6. New tasks via invention and reengineering

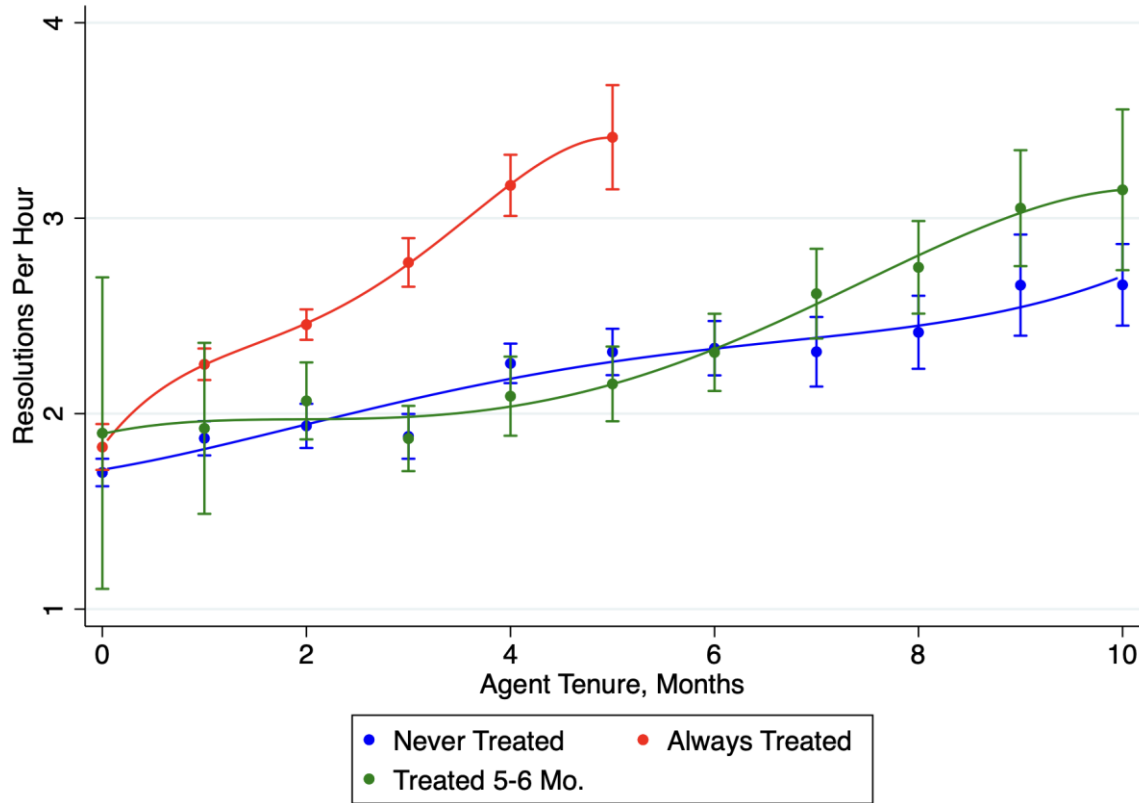
Source: Brynjolfsson and Mitchell “What can Machines Learn?” *Science*, 2017

Where might we see the
largest business value?

There are five big areas where this wave of AI can have the greatest process impact

- Sales
- Marketing
- Customer Service
- Software development
- Analysis

Let's dive in on customer service. Field studies show large effects.

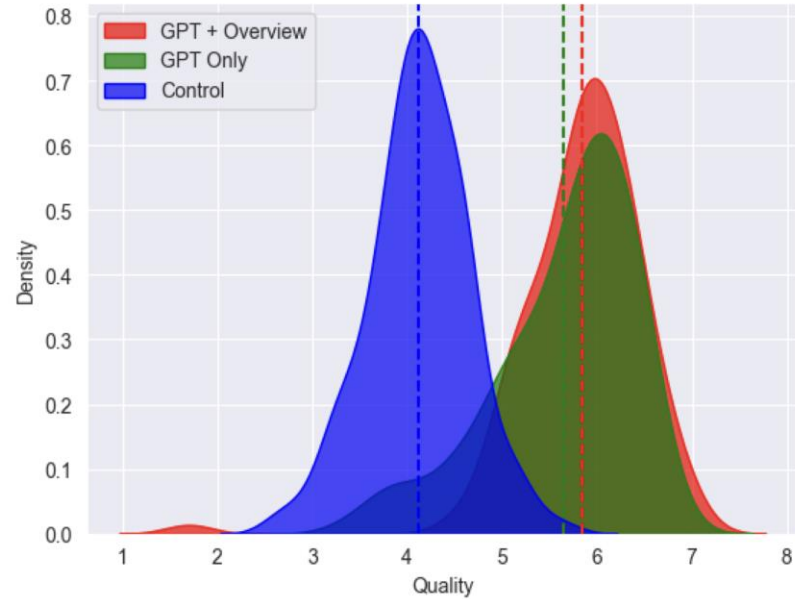


Brynjolfsson, Li, and Raymond (2023)

Similarly it's the case for Analysis... just take a look at BCG!

- 758 Consultants
- Two task groups: develop new products and business problem-solving
- Quality of output evaluated with/without GPT

Figure 2: Performance Distribution - Inside the Frontier

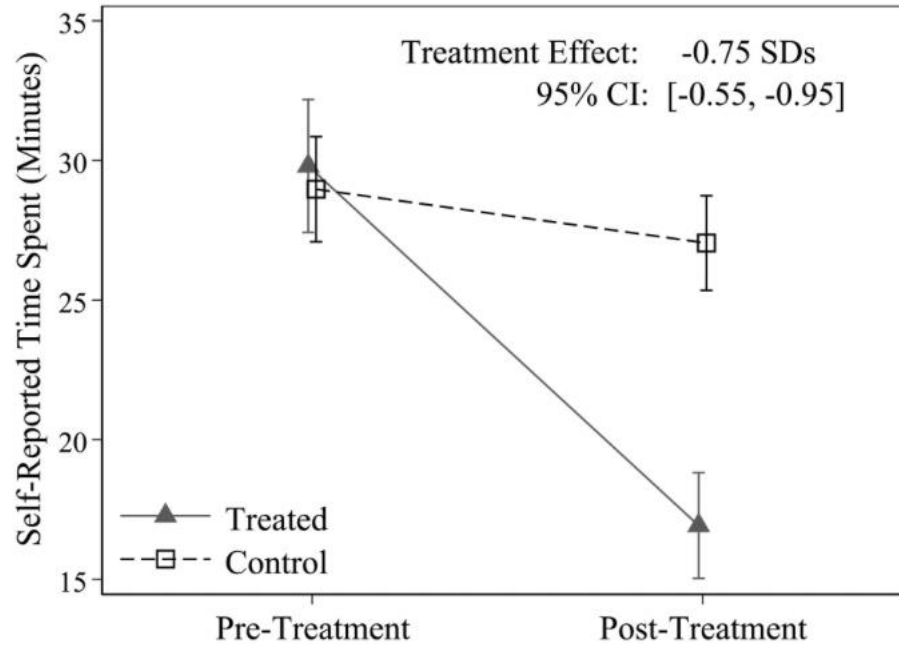


Notes: This figure displays the full distribution of performance in the experimental task inside the frontier for subjects in the three experimental groups (red for subjects in the GPT+Overview condition; green for subjects in the GPT Only condition; blue for subjects in the control condition).

Dell'Acqua et al. (2023)

Or writers... they speed up

A Time Taken Decreases



Noy and Zhang (2023)

Combining supervised learning with generative AI to generate new products



FIGURE 1. HIGH-LEVEL GOAL OF THIS WORK, PREDICTING POTENTIALLY UNKNOWN “DESIGN GAPS” IN THE MARKET.



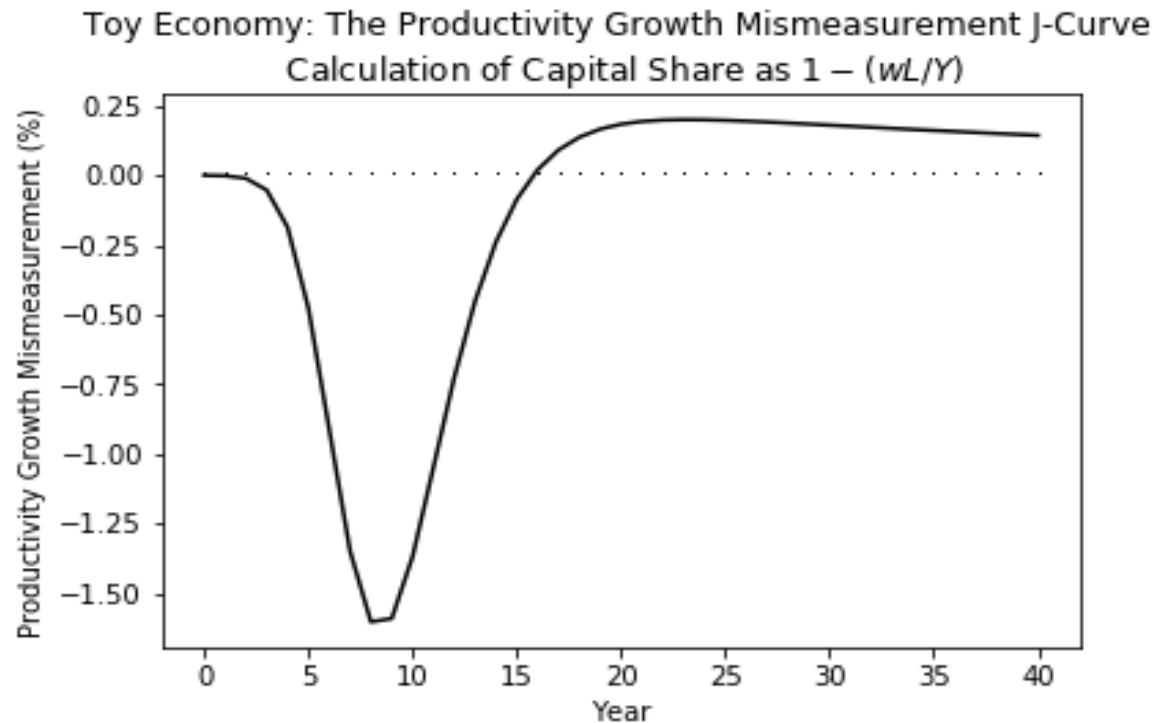
FIGURE 6. ROTATIONS OF VARIOUS BODY TYPES FROM THE ESTIMATED PRODUCT FORM DESIGN SPACE.

The risks are real. Safeguards are necessary.

- Hallucinations/Confabulations: consider these models as clever, but error-prone. What if you had 200 new graduates working for everyone?
- Exploits are real (Air Canada, car dealership)
- Discovery of cybersecurity vulnerabilities.... Social engineering at scale.
- Bias

When Air Canada's chatbot gave incorrect information to a traveller, the airline argued its chatbot is "responsible for its own actions".

Transformational change will take time: The Productivity J-Curve



Some tips and tricks

Different types of AI work together. We're all systems engineers now.

- We can use LLMs to generate training data for supervised models
- Then train an ordinary machine learning model. You've stripped out a module!
- Prompting... what works? CoT, Few-shot vs. zero-shot, asking for explanations
- Ask for the impossible: QA systems? Summaries? New ideas?
- **The main blocker for using AI is training.** Need to give people time.
- Every programmer should use a tool like copilot.
- Organizationally, let people experiment (be careful with data and IP protections)
- Polanyi's paradox and "reverse" Polanyi's paradox. Embrace the latent space.

Returning to the key takeaways: generative AI will be transformative and will require a plan.

- This is not the robot apocalypse. It is a massive opportunity.
- Now is **the time to get moving**. Starting an implementation plan is table stakes because this technology is moving quickly.
- Of course, “**measure twice, cut once**”. It can be helpful to get some quick wins under your belt first.
- **Set up your platform for success**. What assets do you have now?
- **Data, Talent, and Computation** will take you further now than they have in the past. Make these resources accessible for building new projects.
- Given wide-ranging exposure of knowledge work, **workforce strategy will be critical** in determining how successful generative AI projects will be.